

GKE Practice Questions

8 scenario-based questions aligned with the Google Cloud Professional Cloud Architect exam

Topics: cluster modes · availability · private clusters · Ingress · Deployments · PV, PVC and StorageClasses

Question 1

Cluster mode

A startup is moving its containerized microservices to Google Cloud. The team is small, has no dedicated Kubernetes operators, and wants to pay only for the CPU and memory their workloads request rather than for idle VM capacity. Which option best meets these requirements?

A GKE Standard cluster with a single node pool of e2-standard-4 VMs

B GKE Autopilot cluster

C Self-managed Kubernetes on Compute Engine VMs

D GKE Standard cluster with Spot VMs and Cluster Autoscaler

Question 1: Answer

B

CORRECT ANSWER

GKE Autopilot cluster

Why

Autopilot has Google manage nodes, scaling and security, and bills per Pod resource request. Standard mode bills per node VM and needs you to manage node pools, even with Spot VMs. Self-managed Kubernetes adds the most operational work.

Remember for the exam

Autopilot: Google manages nodes, billed per Pod request. Standard: you manage node pools, billed per VM.

Question 2

Availability

A retail company runs its checkout service on GKE. A recent outage in a single zone made the cluster’s control plane unavailable, blocking deployments during a sale. Leadership wants the control plane and workloads to survive a zone failure. What should the architect recommend?

A A zonal cluster with node pools spread across three zones (multi-zonal)

B A regional cluster

C A private zonal cluster

D Two single-zone clusters with a manual failover runbook

Question 2: Answer

B

CORRECT ANSWER

A regional cluster

Why

A regional cluster replicates the control plane and nodes across multiple zones in a region, so a single zone failure doesn’t take down the API server. A multi-zonal cluster spreads nodes, but its control plane is still in one zone.

Remember for the exam

Zonal and multi-zonal clusters have one control plane. Regional clusters replicate it across zones.

Question 3

Networking

A financial services company has a security policy that no compute resource processing customer data may have a public IP address. They are deploying a new application on GKE. Which cluster configuration meets this policy?

A A public cluster with firewall rules blocking all ingress traffic

B A private cluster, so nodes get internal IP addresses only

C A Standard cluster with Shielded GKE nodes enabled

D A cluster that uses only ClusterIP Services

Question 3: Answer

B

CORRECT ANSWER

A private cluster, so nodes get internal IP addresses only

Why

In a private cluster, nodes have internal IPs only and access to the API endpoint can be restricted. Firewall rules or Shielded nodes improve security but leave public IPs on the nodes, and Service type doesn’t change node addressing.

Remember for the exam

Private clusters keep nodes off the public internet; pair with Cloud NAT for outbound access.

Question 4

Services and Ingress

A media company runs two Deployments on GKE: a frontend and an API. They want a single public HTTPS endpoint that routes example.com/ to the frontend and example.com/api to the API, using a Google-managed load balancer. What should they configure?

A Two Services of type LoadBalancer, one per Deployment

B Two NodePort Services and a DNS round-robin record

C An Ingress resource backed by Services for each Deployment

D An ExternalName Service pointing at example.com

Question 4: Answer

C

CORRECT ANSWER

An Ingress resource backed by Services for each Deployment

Why

Ingress on GKE provisions an HTTP(S) Application Load Balancer that routes by URL path to different backend Services through one endpoint. LoadBalancer Services each get their own external IP and don’t route by path; NodePort is mainly for testing; ExternalName is only a DNS alias.

Remember for the exam

LoadBalancer Service: one external IP per Service. Ingress: HTTP(S) routing by host and path.

Question 5

Deployments

A team updated the container image in their GKE Deployment. Within minutes, error rates spiked because of a bug in the new version. They need to restore the previous working version as fast as possible with minimal manual effort. What should they do?

A Delete the Deployment and recreate it from the old YAML file

B Run kubectl rollout undo on the Deployment

C Scale the Deployment to zero replicas, then back up

D Delete all the Pods so the ReplicaSet recreates them

Question 5: Answer

B

CORRECT ANSWER

Run kubectl rollout undo on the Deployment

Why

Deployments keep revision history, and kubectl rollout undo reverts to the previous revision with a rolling update, avoiding downtime. Recreating the Deployment causes an outage, and scaling or deleting Pods just recreates the same broken version.

Remember for the exam

kubectl rollout history lists revisions; kubectl rollout undo reverts to the previous one.

Question 6

Storage access modes

A content management system on GKE runs multiple web server Pods spread across several nodes. All of them must read and write the same set of user-uploaded media files at the same time. Which storage approach meets this requirement?

A A PVC with ReadWriteOnce access backed by a Balanced Persistent Disk

B A PVC with ReadWriteMany access backed by Filestore

C An emptyDir volume in each Pod

D A PVC with ReadWriteOncePod access using the standard-rwo StorageClass

Question 6: Answer

B

CORRECT ANSWER

A PVC with ReadWriteMany access backed by Filestore

Why

ReadWriteMany lets a volume be mounted read-write by many nodes, which on GKE needs Filestore. Persistent Disk with ReadWriteOnce attaches to one node only, ReadWriteOncePod limits it to one Pod, and emptyDir isn’t shared or persistent.

Remember for the exam

RWO: one node · ROX: many readers · RWX: many writers (Filestore) · RWOP: one Pod.

Question 7

Dynamic provisioning

On a regional GKE cluster, a team finds that Pods using dynamically provisioned PVCs sometimes stay Pending because the Persistent Disk was created in a different zone from where the Pod could be scheduled. How should they fix this?

A Pre-create PVs manually in every zone

B Use a StorageClass with volumeBindingMode set to WaitForFirstConsumer

C Change the reclaim policy to Retain

D Switch the cluster from regional to zonal

Question 7: Answer

B

CORRECT ANSWER

Use a StorageClass with volumeBindingMode set to WaitForFirstConsumer

Why

WaitForFirstConsumer delays disk creation until a Pod using the PVC is scheduled, so the disk is created in that Pod’s zone. Pre-creating PVs adds manual work, reclaim policy affects deletion rather than placement, and going zonal gives up high availability.

Remember for the exam

WaitForFirstConsumer creates the disk only after the Pod is scheduled, in that Pod’s zone.

Question 8

Reclaim policy

A healthcare company runs PostgreSQL on GKE using dynamically provisioned storage. A compliance requirement says the underlying disk must not be destroyed if someone accidentally deletes the PersistentVolumeClaim. What should the architect configure?

A Use a StorageClass with reclaimPolicy set to Retain

B Use the default standard-rwo StorageClass with reclaimPolicy Delete

C Set the PVC access mode to ReadOnlyMany

D Run PostgreSQL as a Deployment instead of a StatefulSet

Question 8: Answer

A

CORRECT ANSWER

Use a StorageClass with reclaimPolicy set to Retain

Why

The reclaim policy decides the disk's fate when its claim is deleted. Retain keeps the PV and Persistent Disk so data can be recovered, while Delete (the usual default for dynamic provisioning) removes the disk. Access mode and workload type don't protect the disk.

Remember for the exam

Delete removes the disk with the claim; Retain keeps the PV and disk for manual recovery.

Answer Key

#	Topic	Answer	Correct option
1	Cluster mode	B	GKE Autopilot cluster
2	Availability	B	A regional cluster
3	Networking	B	A private cluster, so nodes get internal IP addresses only
4	Services and Ingress	C	An Ingress resource backed by Services for each Deployment
5	Deployments	B	Run kubectl rollout undo on the Deployment
6	Storage access modes	B	A PVC with ReadWriteMany access backed by Filestore
7	Dynamic provisioning	B	Use a StorageClass with volumeBindingMode set to WaitForFirstConsumer
8	Reclaim policy	A	Use a StorageClass with reclaimPolicy set to Retain



Questions?

Review the answer slides and the GKE deck before your next practice exam